

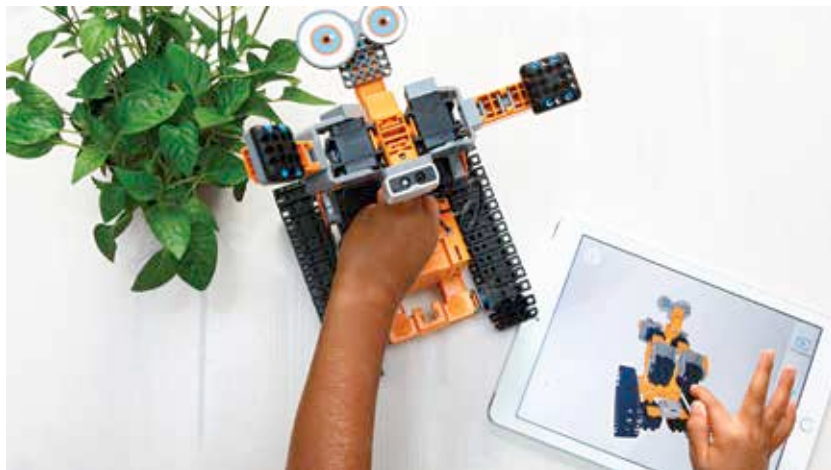
**Playshifu**

Playshifu makes learning fun with engaging AR based applications, as well as phygital board games. <https://digit.in/jul20-05>

**Smartivity**

Smartivity makes cardboard based activity kits for kids. These kits are also available as a subscription. <https://digit.in/jul20-06>

STEM startups push learning through play



Credit: Alesiakian / Shutterstock.com

Embarking on a mission to prepare children for a tech-focused future

Dhriti Datta | dhriti@digit.in

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he inefficiencies of the educational system, especially India's exam-centric system are rather jarring.

The education system inculcates a rather archaic style of learning in young children with the braindead note-taking and rote learning practices that are still extremely prominent not just in India, but many parts of the world. While there is a focus on STEM (science, technology, engineering, and mathematics) subjects in most curriculums, not many tackle these subjects with a practical, hands-on approach which discourages creativity, critical thinking and real-world application. Traditional education systems simply

lack the content and pedagogy that will best prepare children for the tech-focused future we are speedily heading towards. While some curriculums are incorporating STEM-based practical training, some are still miles away. This is where new-age parents and even children are turning towards the marketplace to seek activities and toys that can help them imbibe the teachings of STEM along with the values of tinkering, exploring and learning through play.

STEM INDUSTRY TAKING THE WORLD BY STORM

A study by the National Science Foundation revealed that a whopping 80 percent of jobs created in the next decade will require math and science skills. While the idea of helping children learn during play is definitely not new, the inculcation of STEM in this approach is definitely a new trend. Studies have clearly shown that starting to imbibe STEM learning early on in a child's life reduces the chance of having a fear of numbers. Furthermore, STEM toys also help children by adding spatial awareness, improving

their patience and focusing abilities, boost imagination and creativity, and also help with problem solving.

The drive to play, especially as kids, is inherent. Therefore, mixing up learning and play proves to be a very successful educational model since it capitalizes on the natural instincts of a child. Furthermore, playing doesn't come with as many rules and limitations as regular educational discourse does, which further opens up a child's mind, urging them to find solutions beyond the conventional methods. Unlike traditional education, there is no right method, no grades to worry about and no rote learning.

The growing demand for STEM-based toys and activities is a good indicator of how popular the model of education and play has gotten. The global market for STEM educational toys is valued at a whopping \$29 billion, out of which India accounts for about \$235 million. According to a report by Technavio, the STEM toys market will grow by over \$473 million during 2020-2024, at a CAGR of over 2 percent. These numbers were adjusted post the impact of the COVID-19 pandemic, and the previous estimates showed a CAGR of over 5 percent. Additionally, 35 percent of this growth will come directly from the Asia-Pacific region. This has sparked a rising number of ed-tech startups that specially focus on creating innovative STEM toys and activities that captivate the minds of the future. AI-powered STEM toys are also increasingly gaining prominence in the market, especially amongst working parents. Despite the slight hurdle due to the ongoing pandemic, the STEM toys market is one of the few categories in the toy industry that has consistently grown over the past five to seven years.